Lecture 7: Generative Learning Algorithms

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October 20, 2026



1. Review of Last Week
   - Exponential Families
   - Generalized Linear Models

2. Generative Learning Algorithms
   - Discriminative vs. Generative Learning Algorithms
   - Gaussian Discriminant Analysis (GDA)
   - Linear Discriminant Analysis (LDA)
   - LDA and Logistic Regression
   - MLE for GDA
   - Naïve Bayes

Reading materials & Reference

Reading materials:

- Chapter 4, Stanford CS 229 Lecture Notes, `https://cs229.stanford.edu/notes2022fall/main_notes.pdf`
- Chapter 9, Probabilistic Machine Learning: an introduction.
- Learning From Data Lecture 4: Generative Learning Algorithms, `http://yangli-feasibility.com/`

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Logistic Regression and Exponential Families

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μ is the probability/mean; η is the log-odds. The canonical link is $g(\mu) = \eta = z$.

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Next question: which parts of this probability formula do we keep fixed, and which part do we learn?

Exponential family — four pieces of one distribution

Use the same scalar template as last lecture:

$$p(y;\eta) = \underbrace{h(y)}_{\text{base measure}} \exp\left[\underbrace{\eta\phi(y)}_{\text{parameter–data interaction}} - \underbrace{A(\eta)}_{\text{log-normalizer}}\right].$$

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$$Z(\eta) = \int h(y) e^{\eta\phi(y)}\, dy, \qquad A(\eta) = \log Z(\eta).$$

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Vector extension: replace $\eta\phi(y)$ by $\boldsymbol{\eta}^\top \boldsymbol{\phi}(y)$; h and the normalization rule have the same roles.

Bernoulli: where do the two weights come from?

Same Bernoulli distribution, written with the natural parameter η:

$$p(y;\eta) = \exp[\eta y - A(\eta)] = \frac{e^{\eta y}}{Z(\eta)}, \qquad y \in \{0,1\}.$$

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Substitute each possible outcome:

Outcome $y = 0$

Unnormalized weight: $e^{\eta \cdot 0} = 1$.

$$p(0;\eta) = \frac{1}{Z(\eta)}.$$

Outcome $y = 1$

Unnormalized weight: $e^{\eta \cdot 1} = e^{\eta}$.

$$p(1;\eta) = \frac{e^{\eta}}{Z(\eta)}.$$

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$$p(1;\eta) = \frac{e^{\eta}}{Z(\eta)}.$$

$$p(0;\eta) + p(1;\eta) = \frac{1 + e^{\eta}}{Z(\eta)} = 1 \quad \Longrightarrow \quad Z(\eta) = 1 + e^{\eta}.$$

The rule: sum the weights to get Z, then divide each weight by Z.

Why does the first derivative give the mean?

Recall Bernoulli: $Z(\eta) = 1 + e^{\eta}$ and $A(\eta) = \log Z(\eta)$.

Two possible outcomes

y	Weight	Probability
0	1	$1/Z$
1	e^{η}	e^{η}/Z

Divide each weight by Z.

Differentiate the log total

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$$A'(\eta) = \frac{Z'(\eta)}{Z(\eta)} = \frac{0 \cdot 1 + 1 \cdot e^{\eta}}{Z(\eta)}.$$

Differentiating $e^{\eta y}$ brings down the outcome y.

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Read the normalized weights

$$A'(\eta) = 0 \cdot p(0;\eta) + 1 \cdot p(1;\eta).$$

$$= \mathbb{E}_{\eta}[Y] = \mu.$$

Why the mean? The derivative produces outcome $\times$ weight; division by Z turns this into outcome $\times$ probability.

Why does the second derivative give the variance?

From the previous slide: $A' = Z'/Z = \mu$. Differentiate this quotient once more.

Differentiate the weights again

$Z''(\eta) = 0^2 \cdot 1 + 1^2 \cdot e^{\eta}$.

Differentiate the normalized mean

$$A''(\eta) = \frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2.$$

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Divide by the same total weight:

$$\frac{Z''}{Z} = 0^2 p(0;\eta) + 1^2 p(1;\eta).$$

$$= \mathbb{E}_{\eta}[Y^2].$$

Differentiate the normalized mean

$$A''(\eta) = \frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2.$$

$$= \mathbb{E}_{\eta}[Y^2] - \mu^2.$$

$$= \mathrm{Var}_{\eta}(Y).$$

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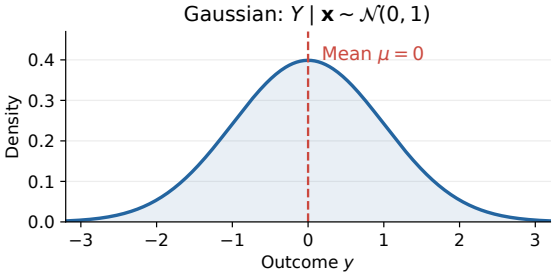
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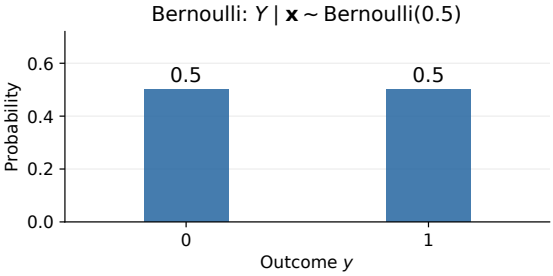
Bernoulli check: $Y^2 = Y \;\Rightarrow\; A''(\eta) = \mu - \mu^2 = \mu(1-\mu)$.

Next question: how do we make the mean μ depend on an input $\mathbf{x}$?

One score, two different predictions

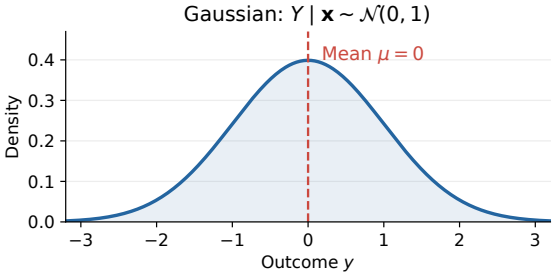
If the score is $z = 0$, must the predicted mean also be zero?

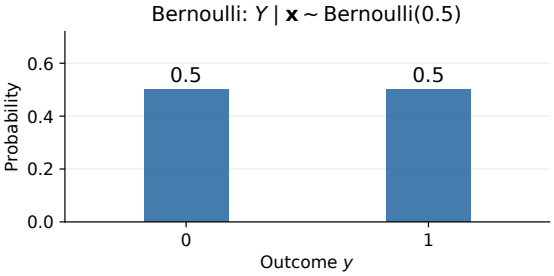




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The family matters: Gaussian gives $\mu = z = 0$; Bernoulli gives $\mu = \sigma(z) = 0.5$.

Next: connect the real-valued score to the mean allowed by each family.

Generalized Linear Models: family, score, and link

Predict the conditional mean $\mu(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$.

Gaussian: real-valued response

$Y \mid \mathbf{x} \sim \mathcal{N}(\mu(\mathbf{x}), 1)$.

Bernoulli: binary response

$Y \mid \mathbf{x} \sim \text{Bernoulli}(\mu(\mathbf{x}))$.

Question: how can a real-valued score produce a valid mean for each family?

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Shared score $z = \mathbf{x}^\top \mathbf{w} + b \in \mathbb{R}$. **Link** $g(\mu) = z$.

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Two directions: $g(\mu) = z$ and $\mu = g^{-1}(z)$.

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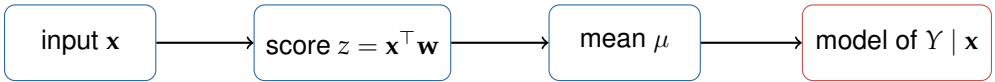
Mean $\mu = \sigma(z)$.

Linear: $g(\mu(\mathbf{x})) = \mathbf{x}^\top \mathbf{w} + b$.

Generalized: the family and link adapt to the response; μ can be nonlinear in z.

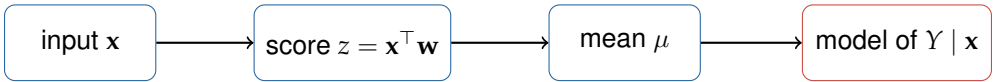
Constructing GLMs: connect the two mappings

Use $\phi(y) = y$; a constant feature absorbs the intercept: $z = \mathbf{x}^\top \mathbf{w}$.



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Distribution family

From natural parameter to mean:

$\eta \longrightarrow \mu = A'(\eta).$

Chosen mean model

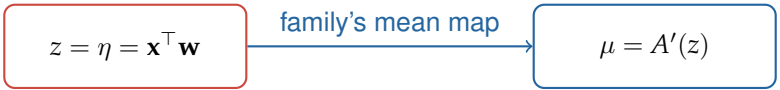
From linear score to mean:

$z \longrightarrow \mu = g^{-1}(z).$

Question: when can we use the score z as the natural parameter η?

Canonical link: let the score be the natural parameter

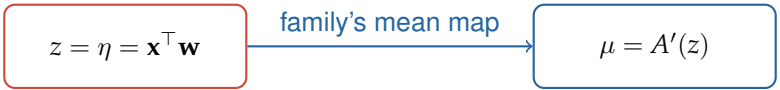
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The conditional model is now fixed:

$$p(y \mid \mathbf{x}; \mathbf{w}) = h(y) \exp[zy - A(z)].$$

Next: learn $\mathbf{w}$ by making the observed outputs likely.

Negative log-likelihood: one observed output

Canonical scalar GLM: $z_n = \mathbf{x}_n^\top \mathbf{w}$, $\eta_n = z_n$, and $\phi(y) = y$.

Probability assigned to the observation

$$p_n = p(y_n \mid \mathbf{x}_n; \mathbf{w}) = h(y_n) \exp[z_n y_n - A(z_n)].$$

take $-\log$

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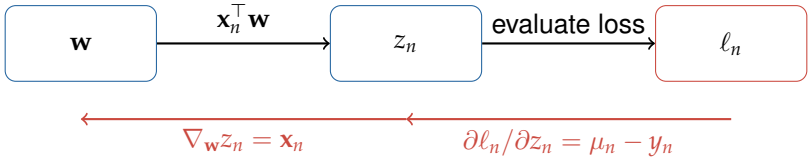
Loss for this observation

$$\ell_n = -\log p_n = A(z_n) - z_n y_n - \log h(y_n).$$

$\log h(y_n)$ is independent of the weights. Higher observed probability means lower loss.

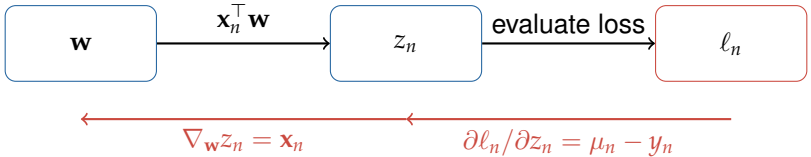
One observation: follow the gradient back to the weights

The model predicts $\mu_n = A'(z_n)$; the observed output is y_n.



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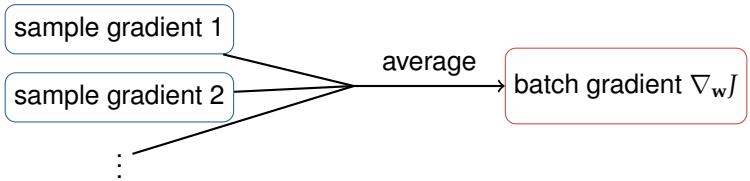


Multiply along the path

$$\nabla_{\mathbf{w}} \ell_n = \underbrace{[A'(z_n) - y_n]}_{\text{predicted mean minus observed output}} \mathbf{x}_n = (\mu_n - y_n)\mathbf{x}_n.$$

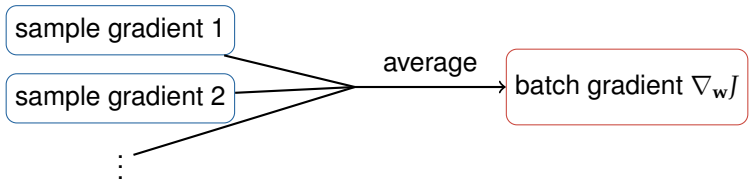
A batch: average the gradients, then update

Independent labels: $L = \prod_n p_n$. Average loss: $J = -\log L/N = \sum_n \ell_n/N$.



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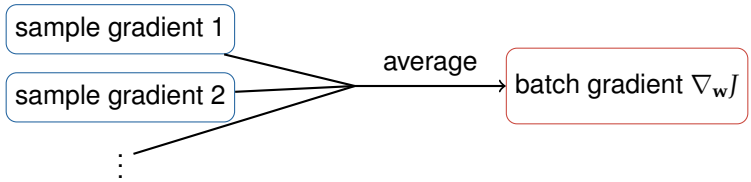


$$\nabla_{\mathbf{w}} J = \frac{1}{N}\sum_n (\mu_n - y_n)\mathbf{x}_n = \frac{1}{N}\mathbf{X}^\top(\boldsymbol{\mu} - \mathbf{y}).$$

$\mathbf{X}$ has rows $\mathbf{x}_n^\top$. For Bernoulli, $\boldsymbol{\mu} = \sigma(\mathbf{X}\mathbf{w})$.

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Update: $\mathbf{w} \leftarrow \mathbf{w} - \alpha\nabla_{\mathbf{w}} J$.

Explicit intercept: $\partial J/\partial b = \sum_n (\mu_n - y_n)/N$.

L2 adds $\lambda\mathbf{w}$ to the weight gradient; the intercept is unpenalized.

From fitting a conditional model to modeling each class

So far: given an input, model its label probability directly:

$$p(1 \mid \mathbf{x}; \mathbf{w}) = \sigma(\mathbf{x}^\top \mathbf{w}).$$

From fitting a conditional model to modeling each class

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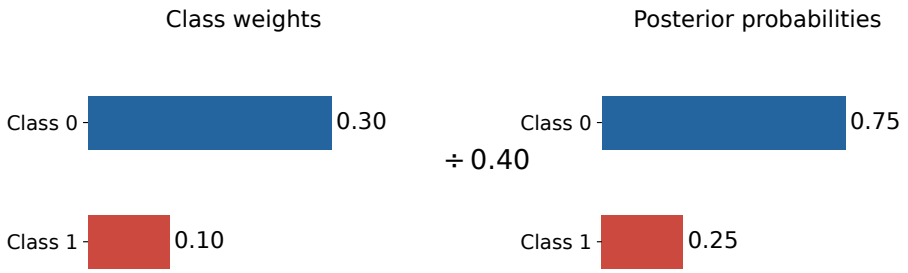
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> Why *generative*? The fitted joint model describes two sampling steps:
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For classification, reverse the question: we observe $\mathbf{x}$ and infer its class. Next, use Bayes' rule. We keep N observations and d input features.

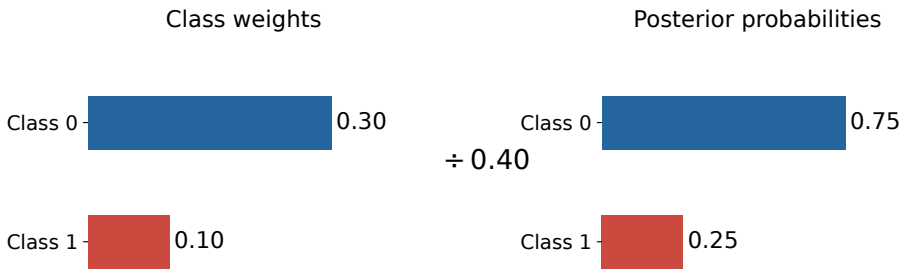
Bayes' rule: rescale the class weights

For one fixed input, each weight already includes its class prior.



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One shared total: $0.30 + 0.10 = 0.40$. Divide both weights by it.

The probabilities sum to 1; the class ordering stays the same.

Recap: the Bayes classifier

Given a new $\mathbf{x}$, we still want the class with the largest posterior (under 0–1 loss):
$\hat{y}(\mathbf{x}) \in \arg\max_c p(y = c \mid \mathbf{x})$.

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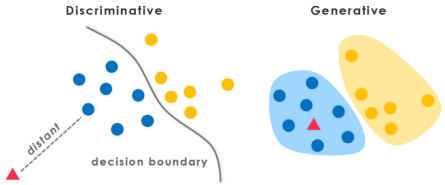
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$p(\mathbf{x} \mid y = c)$ describes inputs within a known class.
$p(y = c \mid \mathbf{x})$ compares classes for a known input. Bayes' rule connects them.

Discriminative vs. Generative Learning Algorithms

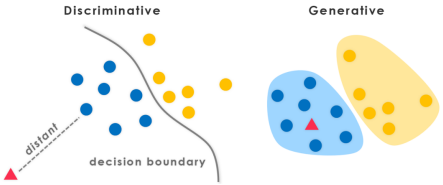


Both methods train on the same observed pairs $(\mathbf{x}_n, y_n)$.

Discriminative: fit the conditional

Generative: fit the joint

Discriminative vs. Generative Learning Algorithms



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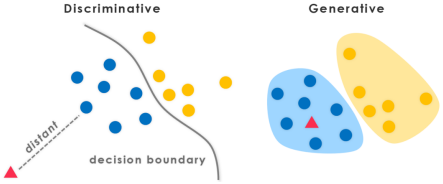
Reward the probability assigned to the observed label at each input.

Generative: fit the joint

$$L(\theta) = \prod_n \pi_{y_n} p(\mathbf{x}_n \mid y_n; \theta).$$

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At prediction time: both compare $p(y = c \mid \mathbf{x})$. The generative model obtains it from the fitted class models using Bayes' rule.

Generative prediction: what cancels?

For one fixed input with $p(\mathbf{x}; \theta) > 0$,

$$p(y = c \mid \mathbf{x}; \theta) = \frac{\pi_c p(\mathbf{x} \mid y = c; \theta)}{p(\mathbf{x}; \theta)}.$$

Question: which parts change as we compare candidate classes?

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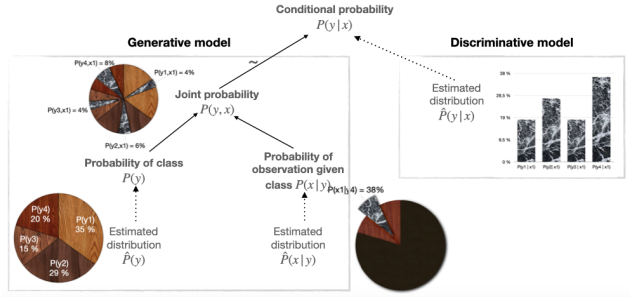
$$\hat{y}(\mathbf{x}) \in \arg\max_c \left[\log p(\mathbf{x} \mid y = c; \theta) + \log \pi_c \right].$$

Interpretation: compare input match plus prior on the log scale. For two classes with equal class-conditional densities at $\mathbf{x}$, the larger prior wins.

Label: compare weights. **Posterior probability:** normalize the weights.

Discriminative vs. Generative: read the same example

Read the two directions: which distributions are fitted before predicting the class?

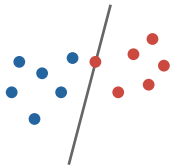


Left: learn an input distribution for each class, then combine it with the prior to obtain $p(y \mid \mathbf{x})$.

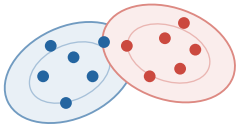
Right: learn $p(y \mid \mathbf{x})$ directly. The output is already the quantity used for classification.

From https://towardsdatascience.com/introduction-to-generative-and-discriminative-models-9c9ef152b9af/

Discriminative vs. Generative: what is assumed?

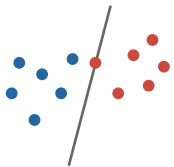


Logistic regression



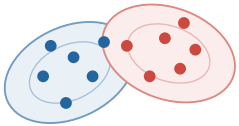
Gaussian class models

Discriminative vs. Generative: what is assumed?



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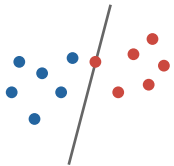
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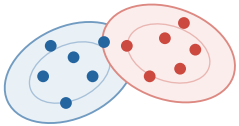
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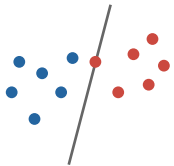


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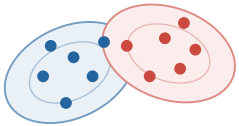


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Not required.



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Required.

Table of Contents

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- LDA and Logistic Regression
- MLE for GDA
- Naïve Bayes

Gaussian Discriminant Analysis (GDA)

We now need a probability model for the inputs within each class. Recall the Gaussian from last lecture:

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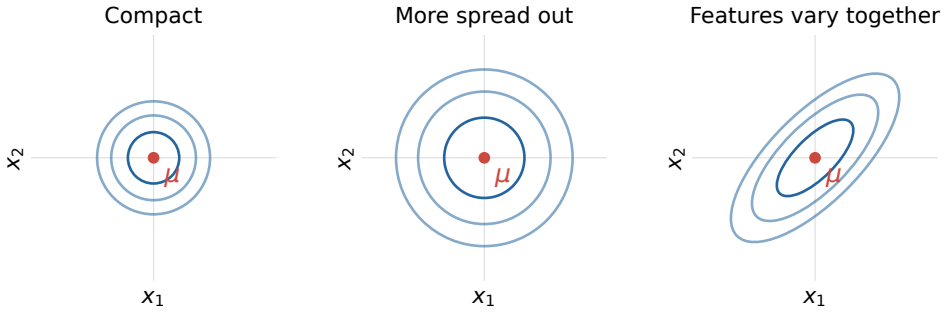
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Same Gaussian family; we now model $\mathbf{x} \mid Y = c$ to predict a class label.

Next question: what changes when a Gaussian observation is a vector?

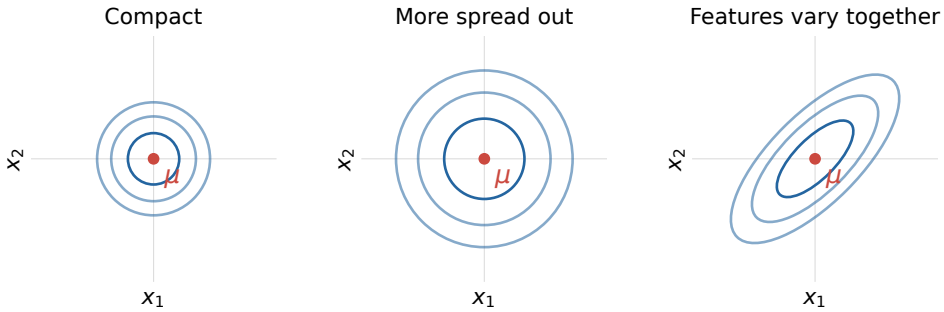
Read a Gaussian as a contour map

Each contour connects inputs with the same density within that panel.



Read a Gaussian as a contour map

Each contour connects inputs with the same density within that panel.



Mean μ: center of the contours. **Covariance Σ:** spread and orientation.

Next: read these two roles in the multivariate Gaussian formula.

The multivariate normal distribution

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{\exp[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})]}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}}.$$

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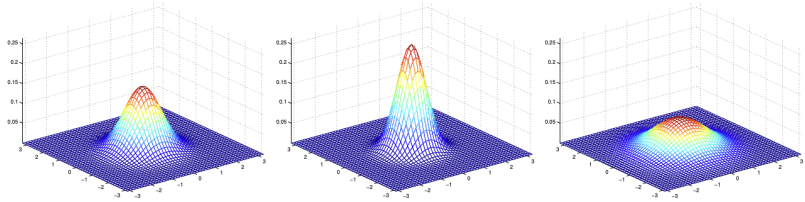
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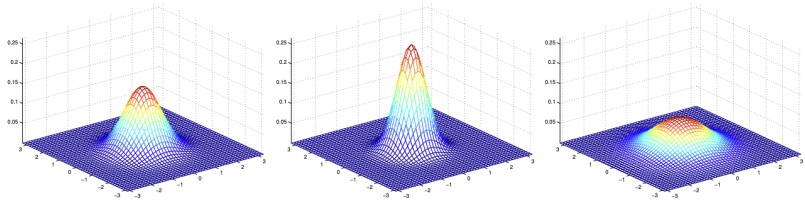


Left: $\boldsymbol{\Sigma} = \mathbf{I}$; middle: $0.6\mathbf{I}$; right: $2\mathbf{I}$. Larger isotropic variance gives a more spread-out density.

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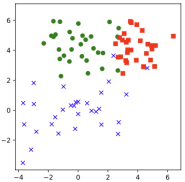


Left: $\boldsymbol{\Sigma} = \mathbf{I}$; middle: $0.6\mathbf{I}$; right: $2\mathbf{I}$. Larger isotropic variance gives a more spread-out density. A covariance may be semidefinite in general; a singular Gaussian has no density of the above form.

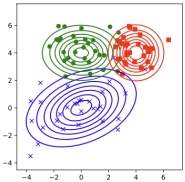
Gaussian Discriminant Analysis (GDA) Model

For C classes, let $\pi_c > 0$, $\sum_c \pi_c = 1$, and

$$Y \sim \text{Categorical}(\pi_1, \ldots, \pi_C), \qquad \mathbf{x} \mid Y = c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c).$$



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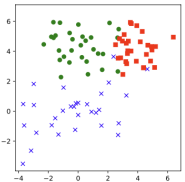


(b)

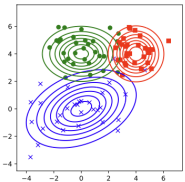
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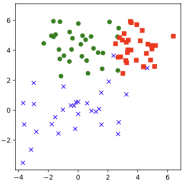
(b)

For binary labels $Y \in \{0, 1\}$, write $Y \sim \text{Bernoulli}(\pi_1)$, with $\pi_0 = 1 - \pi_1$.

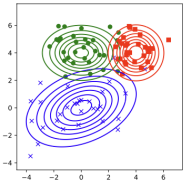
Gaussian Discriminant Analysis (GDA) Model

For C classes, let $\pi_c > 0$, $\sum_c \pi_c = 1$, and

$$Y \sim \text{Categorical}(\pi_1, \ldots, \pi_C), \qquad \mathbf{x} \mid Y = c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c).$$



(a)



(b)

For binary labels $Y \in \{0, 1\}$, write $Y \sim \text{Bernoulli}(\pi_1)$, with $\pi_0 = 1 - \pi_1$.

Two choices: class-specific covariance $\boldsymbol{\Sigma}_c$ (QDA), or shared covariance $\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$ (LDA).

GDA Model: from densities to class scores

As in the GLM likelihood, taking a log turns products into sums. For each class, define

$$s_c(\mathbf{x}) := \log \left[\pi_c \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c) \right]$$

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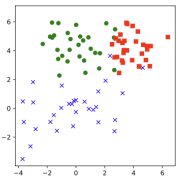
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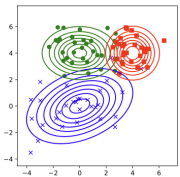
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What stays the same across classes? The last term is common; the log-determinant term is common when covariances are shared. Next, compare the boundaries with and without sharing.

GDA geometry: separate or shared covariance?



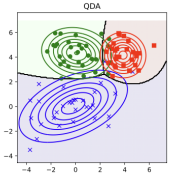
(a)



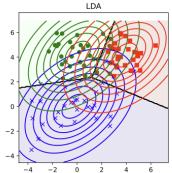
(b)

Class-specific Σ_c: QDA

Pairwise boundaries are generally quadratic.



(a)

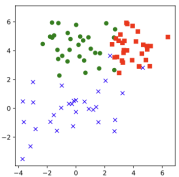


(b)

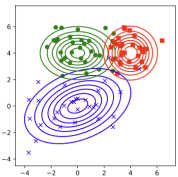
Shared Σ: LDA

Pairwise boundaries are linear.

GDA geometry: separate or shared covariance?

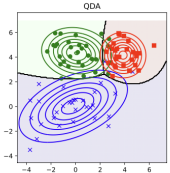


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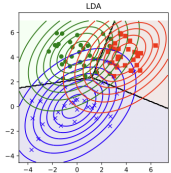


(b)

Class-specific Σ_c: QDA
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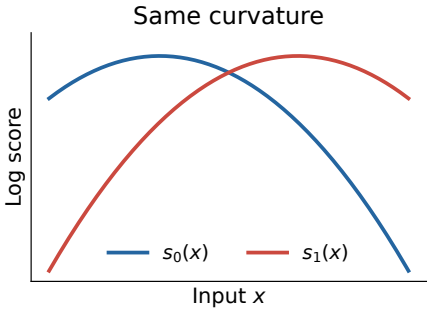
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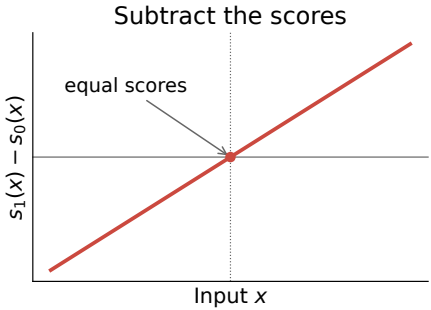
Shared Σ: LDA
Pairwise boundaries are linear.

Next question: which quadratic terms cancel when $\Sigma_c = \Sigma$?

Shared covariance: the curvature cancels

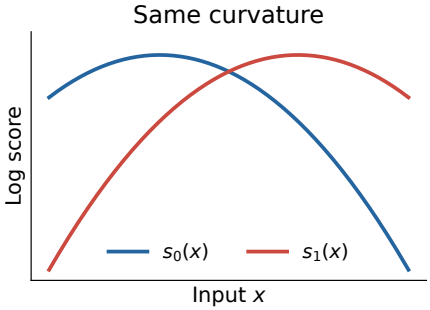
Look at one input dimension, with equal priors and a shared variance.

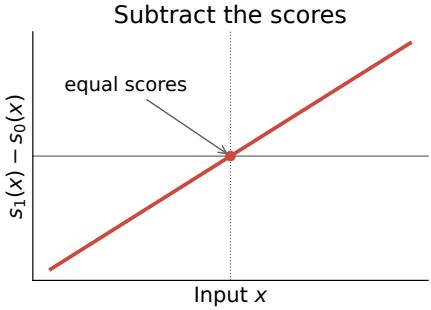




Shared covariance: the curvature cancels

Look at one input dimension, with equal priors and a shared variance.





Two curved log scores can have a **straight-line difference**.

Decision boundary: where their difference is zero. Next, find the cancelling term.

What is LDA?

LDA shares one covariance matrix across all classes.

Why does this make the decision boundary linear?

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$$(\mathbf{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) = \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{x} - 2\mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c + \boldsymbol{\mu}_c^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c.$$

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The first term depends on $\mathbf{x}$, but **not on the class**. It cancels when comparing classes.

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Linear Discriminant Analysis (LDA)

With $\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}$, expand the Gaussian log score:

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The connection between LDA and logistic regression

Recall $s_c(\mathbf{x}) = \delta_c(\mathbf{x}) + \kappa(\mathbf{x})$, where $\delta_c = \gamma_c + \mathbf{x}^\top \boldsymbol{\beta}_c$.

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Shared-covariance Gaussian class models give a **softmax posterior** with linear scores.

For two classes, softmax reduces to the sigmoid from Lecture 06.

Binary LDA: recover the sigmoid

For two classes, recall the Bernoulli log-odds:

$$\eta(\mathbf{x}) := \log \frac{p(1 \mid \mathbf{x}; \theta)}{p(0 \mid \mathbf{x}; \theta)} = \delta_1(\mathbf{x}) - \delta_0(\mathbf{x}) = \mathbf{x}^\top \mathbf{w} + b,$$

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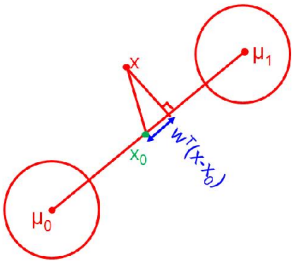
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Next question: where is the decision boundary?

Geometry interpretation of LDA

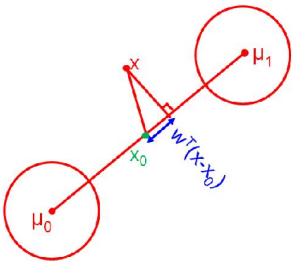
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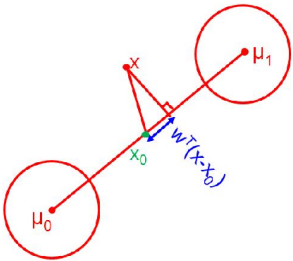
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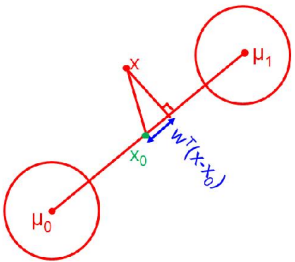
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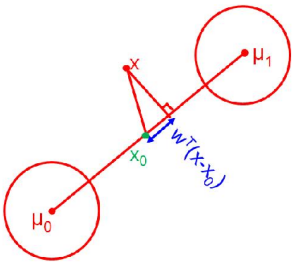
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- With both assumptions, choose the nearest mean in Euclidean distance.
- A general shared covariance changes the direction; unequal priors shift the boundary through b.



LDA by hand: how does the prior move the boundary?

One-dimensional example: $\mu_0 = 0$, $\mu_1 = 2$, shared variance $\sigma^2 = 1$.

$$w = 2, \qquad b = -2 + \log(\pi_1/\pi_0),$$
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- $\pi_1 = 0.8$, $\pi_0 = 0.2$: boundary $x = 1 - \frac{1}{2}\log 4 \approx 0.307$.

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- $\pi_1 = 0.8$, $\pi_0 = 0.2$: boundary $x = 1 - \frac{1}{2}\log 4 \approx 0.307$.
- At $x = 1$, the likelihoods are equal, but the posterior is now 0.8.

LDA by hand: how does the prior move the boundary?

One-dimensional example: $\mu_0 = 0$, $\mu_1 = 2$, shared variance $\sigma^2 = 1$.

$$w = 2, \qquad b = -2 + \log(\pi_1/\pi_0),$$
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Explain: a more common class needs less likelihood evidence to be predicted.

Some remarks: same posterior form, different fitting

We have derived the same sigmoid form. Compare what the two fitting procedures reward:

LDA: fit labeled inputs jointly

$$L(\theta) = \prod_n p(\mathbf{x}_n, y_n; \theta).$$

Fit the class priors and shared-covariance Gaussian class models.

Logistic: fit conditional labels

$$L(\mathbf{w}, b) = \prod_n p(y_n \mid \mathbf{x}_n; \mathbf{w}, b).$$

Fit a sigmoid with linear log-odds directly.

Question: why can the fitted coefficients differ on the same data?

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LDA: fit labeled inputs jointly

$$L(\theta) = \prod_n p(\mathbf{x}_n, y_n; \theta).$$

Fit the class priors and shared-covariance Gaussian class models.

Then derive $\mathbf{w}, b$ from those fitted parameters.

Logistic: fit conditional labels

$$L(\mathbf{w}, b) = \prod_n p(y_n \mid \mathbf{x}_n; \mathbf{w}, b).$$

Fit a sigmoid with linear log-odds directly.

Optimize $\mathbf{w}, b$ using conditional NLL; no model for $p(\mathbf{x})$ is fitted.

Question: why can the fitted coefficients differ on the same data?

Some remarks: same posterior form, different fitting

Expand the joint likelihood using the conditional probability it induces:

$$\begin{aligned}\log L(\theta) &= \sum_n \log p(\mathbf{x}_n, y_n; \theta) \\ &= \underbrace{\sum_n \log p(y_n \mid \mathbf{x}_n; \theta)}_{\text{fit the labels given the inputs}} + \underbrace{\sum_n \log p(\mathbf{x}_n; \theta)}_{\text{also fit the inputs}}.\end{aligned}$$

The same θ controls both terms. Joint fitting must explain the inputs as well as their labels; conditional fitting rewards only the label probabilities.

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The same θ controls both terms. Joint fitting must explain the inputs as well as their labels; conditional fitting rewards only the label probabilities.

The sigmoid describes how the fitted model predicts.

The likelihood defines how its parameters are learned.

Same posterior family does not imply the same fitted coefficients.

Quick check: why does LDA give a sigmoid?

Which statement is correct for two-class LDA?

A. Any two Gaussian class models always give linear log-odds.

B. Shared covariance cancels the quadratic input terms, leaving linear log-odds.

C. LDA and logistic regression maximize the same likelihood.

Think / **vote:** identify the term that cancels.

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Answer: B. The shared term $-\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x}/2$ cancels. Different covariances generally leave quadratic terms.

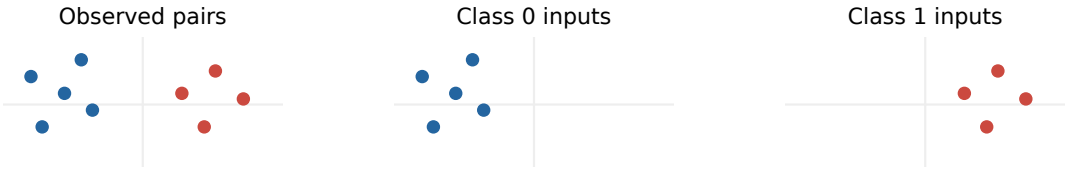
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Fit GDA: the training labels tell us which class to fit

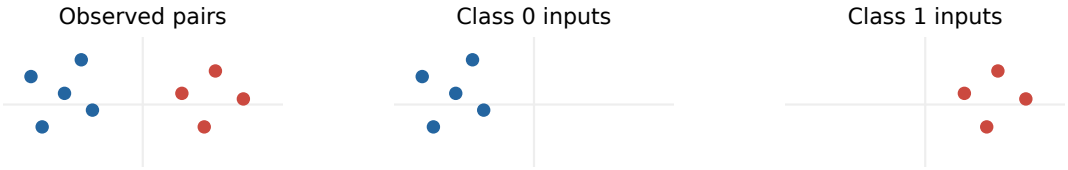
Known labels split the training points into classes; N_c counts the points in class c.



$$L(\theta) = \prod_n \pi_{y_n} \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_{y_n}, \boldsymbol{\Sigma}_{y_n}).$$

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Taking logs groups the observations by class:

$$\log L = \underbrace{\sum_c N_c \log \pi_c}_{\text{class frequencies}} + \underbrace{\sum_c \sum_{n:y_n=c} \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)}_{\text{fit each class's inputs}}.$$

Fit GDA: count, find the center, measure the spread

Let $N_c = \sum_n \mathbf{1}\{y_n = c\} > 0$. For class-specific covariance (QDA):

1. Count

Class frequency.

$$\hat{\pi}_c = \frac{N_c}{N}.$$

2. Center

Average its inputs.

$$\hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{n:y_n=c} \mathbf{x}_n.$$

3. Spread

Center within this class:

$$\mathbf{r}_n = \mathbf{x}_n - \hat{\boldsymbol{\mu}}_c.$$

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Again, match means within each class. The covariance MLE divides by N_c.

The full-dimensional density requires nonsingular scatter. Next, share one covariance.

MLE for LDA: write the shared-covariance objective

Use one covariance Σ for every class. With $\mathbf{r}_n = \mathbf{x}_n - \boldsymbol{\mu}_{y_n}$,

$$\log L(\theta) = \sum_c N_c \log \pi_c - \frac{N}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_n \mathbf{r}_n^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}_n - \frac{Nd}{2} \log(2\pi).$$

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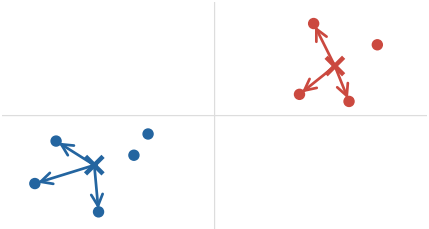
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- The result is the within-class sample mean, just as in the shared-parameter MLE review.

Shared covariance: pool the spreads, not the centers

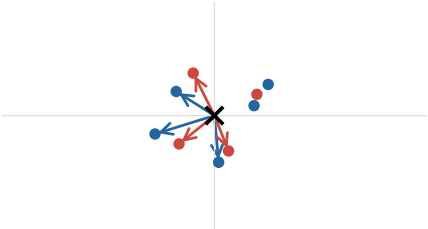
Subtract each point's own class mean before combining the residuals.

Points and their class centers



x_n

Move each class center to zero

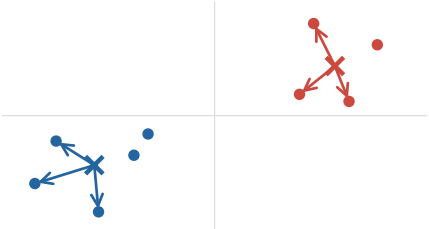


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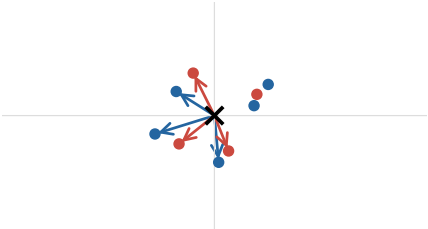
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Each arrow keeps its direction and length when its class center moves to zero.

Then pool: estimate the shared covariance from all these residuals.

MLE for LDA: pool the within-class residuals

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- To obtain $\mathbf{w}$, solve $\hat{\boldsymbol{\Sigma}}_\lambda \mathbf{w} = \hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_0$ rather than forming the matrix inverse explicitly.

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From Gaussian inputs to binary text features

So far: model the inputs within each class, then apply Bayes' rule. For GDA, we chose a Gaussian distribution for those inputs.

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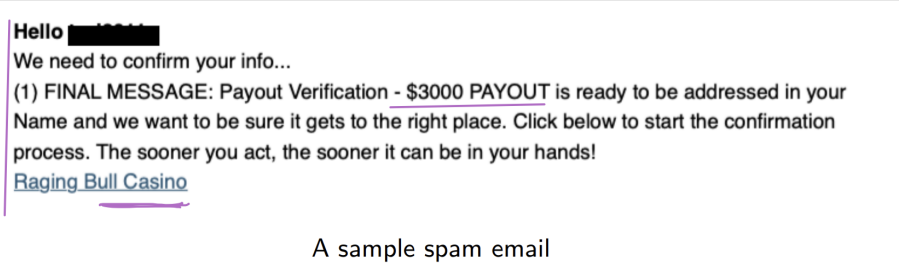
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We will combine these feature models using a conditional independence assumption. This is Bernoulli Naïve Bayes; other Naïve Bayes models can use other feature distributions.

Naïve Bayes: Motivating Example

Example 1 (Spam filter (document classification))

Classify email message $\mathbf{x}$ to spam ($y = 1$) and non-spam ($y = 0$) classes.



Hello [redacted]

We need to confirm your info...

(1) FINAL MESSAGE: Payout Verification - $3000 PAYOUT is ready to be addressed in your Name and we want to be sure it gets to the right place. Click below to start the confirmation process. The sooner you act, the sooner it can be in your hands!

Raging Bull Casino

A sample spam email

Example: Spam Filter

Binary text features: fix a dictionary of d words and represent an email by $\mathbf{x} \in \{0,1\}^d$.

$$x_j = \begin{cases} 1, & \text{word } j \text{ occurs in the email,} \\ 0, & \text{otherwise.} \end{cases}$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \text{a} \\ \text{aardvark} \\ \text{aardwolf} \\ \vdots \\ \text{buy} \\ \vdots \\ \text{zygmurgy} \end{matrix}$$

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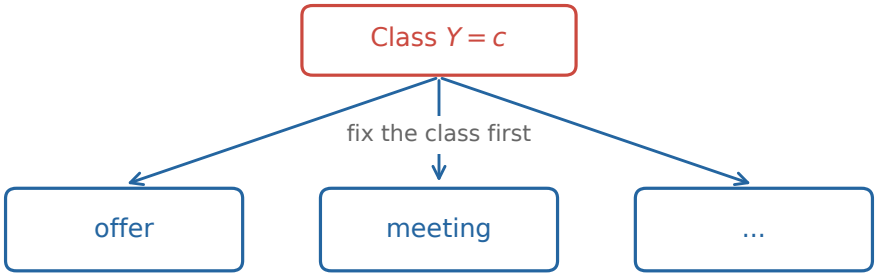
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Learn $p(\mathbf{x} \mid y = c)$ from messages in class c.

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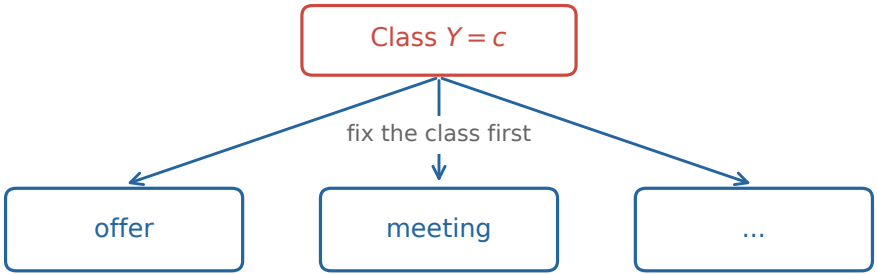
Naïve Bayes: fix the class, then model the words

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Within one class, each word-presence feature has its own Bernoulli model.



The assumption: once the class is fixed, other words do not change a word's probability.

Multiply the word factors *within that class*; repeat for the other class.

Naïve Bayes assumption and classifier

The chain rule always gives

$$p(\mathbf{x} \mid y = c) = \prod_{j=1}^{d} p(x_j \mid x_1, \ldots, x_{j-1}, y = c).$$

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- Real words are related even within a class; this simplifying assumption can still give a useful classifier.

Naïve Bayes Parameters

For binary spam labels $Y \in \{0, 1\}$, let $p(y = c) = \pi_c$ and $q_{jc} := p(x_j = 1 \mid y = c)$, with $0 \leq q_{jc} \leq 1$.

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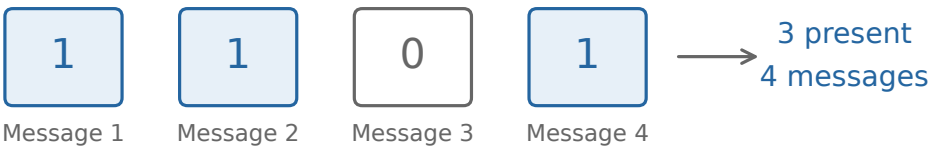
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- One free class prior and $2d$ word-presence probabilities.
- Each q_{jc} is a shared mean for word j in class c. **Next question:** how did we fit a shared Bernoulli mean last lecture?

Fit one word in one class: count the ones

Last lecture: a shared Bernoulli mean is the fraction of observed ones.

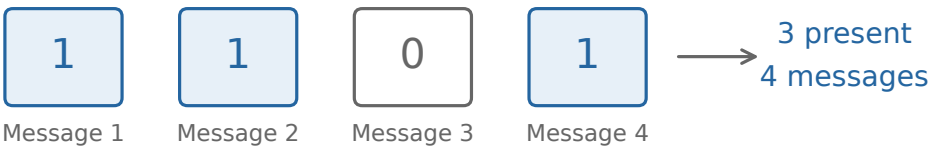
One word j; one fixed class c



Fit one word in one class: count the ones

Last lecture: a shared Bernoulli mean is the fraction of observed ones.

One word *j*; one fixed class *c*



$$\hat{q}_{jc} = \frac{N_{jc}}{N_c} = \frac{1}{N_c} \sum_{n:y_n=c} x_{nj}.$$

N_c: messages in class c. N_{jc}: those containing word j.

Repeat for each word and each class; $\hat{\pi}_c = N_c/N$.

Why counting ones maximizes the Bernoulli likelihood

Each word model shares one q_{jc} across the messages in class c.

$$L(\theta) = \prod_c \left[\pi_c^{N_c} \prod_j q_{jc}^{N_{jc}} (1 - q_{jc})^{N_c - N_{jc}} \right].$$

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Boundary case: all-zero or all-one observations give probability MLE 0 or 1.

Next: use the fitted word probabilities to score a new message.

Score a message: every word selects one factor

For word j, the observed feature x_j chooses which Bernoulli outcome occurred.

Present: $x_j = 1$

Use $\hat{q}_{jc}$.

Log-score contribution: $\log \hat{q}_{jc}$.

Absent: $x_j = 0$

Use $1 - \hat{q}_{jc}$.

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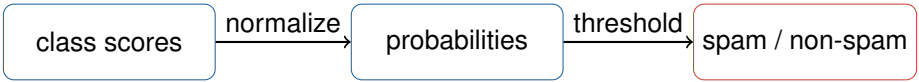
Add the log prior and all word contributions:

$$s_c(\mathbf{x}) = \log \hat{\pi}_c + \sum_{j=1}^{d} \left[x_j \log \hat{q}_{jc} + (1 - x_j) \log(1 - \hat{q}_{jc}) \right].$$

Log scores avoid products of tiny probabilities. Absent words contribute too.

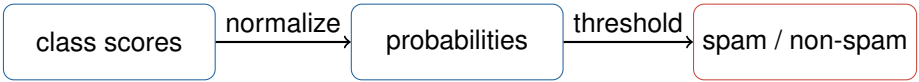
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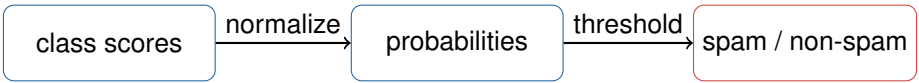


$$p(y = c \mid \mathbf{x}) = \frac{e^{s_c(\mathbf{x})}}{\sum_k e^{s_k(\mathbf{x})}}, \qquad p(1 \mid \mathbf{x}) = \sigma(s_1 - s_0).$$

Default 0–1 decision: spam if $p(1 \mid \mathbf{x}) > 0.5$; ties arbitrary.

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Raising the threshold reduces false positives (blocked legitimate mail) and increases false negatives (missed spam), with scores held fixed.

The log scores require probabilities in $(0, 1)$; smoothing supplies them next.

Naïve Bayes by hand: follow both word factors

The message contains `offer`, but not `meeting`: $\mathbf{x} = (1, 0)$.

	Prior	`offer`	no `meeting`	Weight

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Absent still counts: ignoring `meeting` gives $0.8/(0.8 + 0.1) \approx 0.889$,
a different posterior under a different scoring rule.

Laplace smoothing: when zero counts break prediction

Suppose a dictionary word never occurs in either class in the training set:

$$N_{j0} = N_{j1} = 0 \quad \Longrightarrow \quad \hat{q}_{j0} = \hat{q}_{j1} = 0.$$

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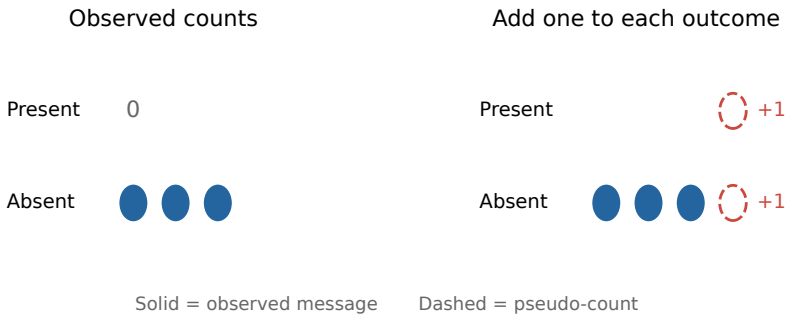
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- A zero in just one class can force an extreme posterior without enough data to justify it.

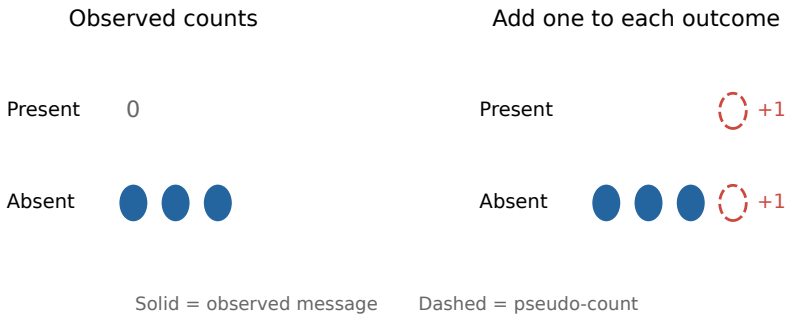
Smoothing: give both outcomes a count

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Presence probability: $\frac{0}{3} = 0 \quad \longrightarrow \quad \frac{0+1}{3+2} = \frac{1}{5}.$

Two outcomes receive a pseudo-count, so the total grows by 2.

Laplace smoothing: add a count to every outcome

For a categorical variable with K possible outcomes and M observations, let M_k be the count of outcome k.

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- This is a smoothed estimate, not the unsmoothed MLE.

Naïve Bayes with Laplace smoothing

Apply the two-outcome rule to word j within class c:

$$\boxed{\tilde{q}_{jc} = \frac{N_{jc} + 1}{N_c + 2}}, \qquad 1 - \tilde{q}_{jc} = \frac{N_c - N_{jc} + 1}{N_c + 2}.$$

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- More generally, add $\alpha > 0$ to both outcomes: $\tilde{q}_{jc} = (N_{jc} + \alpha)/(N_c + 2\alpha)$. Choose any tuned α using validation data.

Quick check: which denominator?

Among three spam messages, none contains `offer`. What is its Bernoulli Naïve Bayes probability after add-one smoothing?

A. $0/3 = 0$.

B. $(0+1)/(3+1) = 1/4$.

C. $(0+1)/(3+2) = 1/5$.

Think / **vote:** how many possible outcomes does this feature have?

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Answer: C. One pseudo-count for presence and one for absence: $\tilde{q} = 1/5$ and $1 - \tilde{q} = 4/5$.

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